Marmara University Electrical-Electronics Eng. Dept.

CSE2003 Digital Design Laboratory Assignment #6

The studies will be uploaded to the system before due date (two weeks). **Each group must present their own study. The group members have to work together and each member must know each step of the study.**

For this laboratory assignment, you will use Altera Quartus II software and DE-115 FPGA kit and design a hardware floating point multiplier that has the Input/output defiitions below.

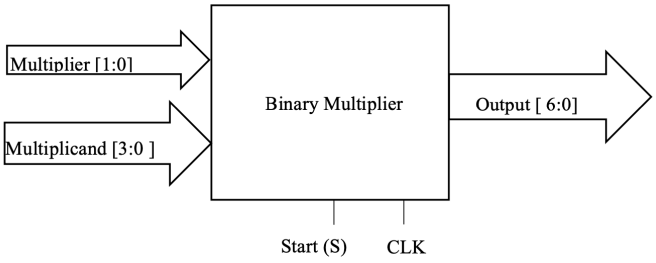
The steps you need to follow: 1. You should design the circuit, **2.**write the verilog HDL code fort he designed circuit, **3.** Check if the designed hardware is correctly coded by using RTL viewer. Don't forget: your circuit may look different then yours since the compiler can make some optimizations. Please take this information into account when you compare the RTL viewer circuit and yours. **4.** Verify your design by using simulation tool (you need to check the inputs and outputs). **5.** implement and RUN your code using Altera DE-115 FPGA kit. Use the remote desktop connection (**Remote desktop connection ID**:256 521 711 **password**: dijitallab2021).

You need to upload the following files before due date:

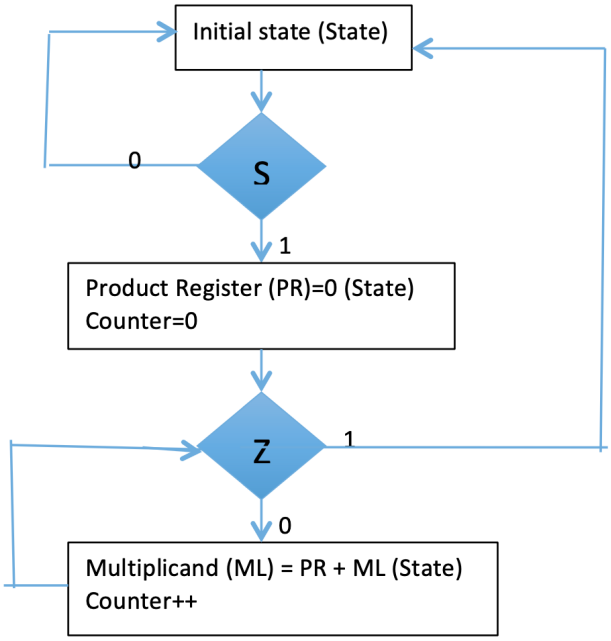
1. A short video (<60 secs.) that shows all steps of the study. Especially, the implementation part must be included.
2. The report (in PDF format): 1. What is the purpose of the study? 2. What are the design details with necessary explenations. 3. The design and implementation steps (all steps with screenshots etc.). 4. The result and coclusion: did you reach the purposed result? If the answer is no, why? if it is yes, how?

For this laboratory assignment, you will use Altera Quartus-II software and DE2-115 FPGA kit and design a binary multiplier that has the Input/output definitions below.

- S signal starts multiplication.
- CLK is the pulse signal which controls the flow in sequential circuit.
- Multiplicand is a 4-bit binary number input.
- Multiplier is 2-bit binary number input.
- Output has 6 bits.



Multiplication will be performed by adding the 'multiplicand' number. 'Multiplier' times addition will give the result. The following chart may help you to understand the required states and signals:



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The Z signal is produced by comparing the Multiplier (ML) and the counter. This signal is logic1 if Counter = ML. otherwise Z=0. Adder circuit should add the ML to the PR until Z=1 is obtained.

1. Determine the subblocks and their inputs/outputs to built the binary multiplier.

2. Design these sub-blocks (i.e.: Adder block,comparator,etc.)
3. Develop the state diagram, and design the FSM that controls the blocks.
4. Design the logic circuit.

5. Implement the circuit using FPGA.

- You are expected to test your device and to demonstrate it by using the output LEDs/7-segments. You may use the switches for input numbers.